PyYield

CMOS Statistical EDA Platform

Complete Variability Analysis & Yield Prediction Report

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Technology Node: 180nm CMOS | Supply: 3.3V | Engine: Ngspice LEVEL-1

Tool	PyYield v2.0 - Open-Source CMOS EDA Platform
Physics	Ngspice LEVEL-1 MOSFET + Analytical CMOS Models
Statistics	Monte Carlo LHS + Pelgrom Local Mismatch (RDF)
ML Model	Random Forest Surrogate + SHAP Explainability
Aging	NBTI + HCI + Electromigration (Black Equation)
Modules	MC, Oracle, Wafer, PVT, Corner, Aging, Power, Tornado
Node	180nm CMOS (configurable: 130nm - 500nm)
Spec Limits	Delay < 0.022 ns	Power < 105.0 uW

Active Simulation Parameters

Supply Voltage	3.3 V
NMOS Vth_n	0.5 V
PMOS Vth_p	-0.5 V
Channel Length	0.18 um
NMOS Width	0.36 um
PMOS Width	0.72 um
Oxide Thickness tox	4.0 nm
Temperature	27 C
Wire Length	5.0 um

Module 1 - Monte Carlo Analysis (LHS + Pelgrom Mismatch)

500 CMOS inverter samples were simulated using Latin Hypercube Sampling with Pelgrom local mismatch (RDF) modeling. Each sample represents one possible fabricated chip under realistic 180nm manufacturing conditions.

Total Chips Simulated	300
Chips Passing Yield	281
Chips Failing Yield	19
Yield Percentage	**93.67%**
Mean Propagation Delay	**0.019028 ns**
Std Dev (Delay)	0.00198 ns
3-Sigma Delay Bound	0.024968 ns
Spec Limit (Delay)	0.022 ns
Mean Power Dissipation	**97.6841 uW**
Std Dev (Power)	12.0226 uW
Spec Limit (Power)	105.0 uW

Module 1B - Process Capability Analysis (Cpk / Cp)

Cp measures potential process capability. Cpk additionally accounts for centering. Industry standard: Cpk >= 1.33 = capable, >= 1.67 = excellent.

Delay Cp	84.162
Delay Cpk	**3.203**
Delay Sigma Level	9.61 sigma
Power Cp	6.931
Power Cpk	**2.708**
Power Sigma Level	8.12 sigma
Process Capable (Cpk >= 1.33)	**YES - Capable**

Module 2 - ML Yield Oracle (Random Forest Surrogate)

The Random Forest surrogate model (200 estimators, trained on Ngspice data) classifies each chip instantly without running new SPICE simulations. SHAP values identify which process parameters most influence the prediction. The Oracle classified the same chip population as the Monte Carlo analysis.

Oracle Yield Estimate	**70.0%**
Chips Classified	300
Passing Classification	210
Failing Classification	90
Predicted Mean Delay	**0.018978 ns**
Predicted Std Delay	0.001732 ns
Predicted Mean Power	**97.8039 uW**
Predicted Std Power	10.8438 uW

SHAP Analysis: Channel length (L) and load capacitance are the dominant contributors to delay prediction. Vth_n and Vth_p contribute secondary effects. Random Forest R-squared > 0.95 on held-out test set.

Module 3 - PVT Corner Analysis (FF / TT / SS)

Corner analysis evaluates circuit performance at three manufacturing extremes. All three corners are simulated using real Ngspice transient analysis with LEVEL-1 MOSFET models. A FAIL indicates the corner exceeds the spec limit.

Fast-Fast	**Delay: 0.015324 ns | Power: 138.798878 uW | PASS**
Low Vth, short L - best-case speed, risk of hold violations	
Typical-Typical	**Delay: 0.019028 ns | Power: 96.864636 uW | PASS**
Nominal parameters - design center and optimization target	
Slow-Slow	**Delay: 0.023955 ns | Power: 75.856225 uW | FAIL**
High Vth, long L - worst-case timing, must meet spec	

Module 4 - Wafer Spatial Yield Analysis (300mm)

The wafer spatial model simulates process variation across a 300mm silicon wafer using a quadratic radial sigma gradient: edge sigma = 3x center sigma. This models the thermal and chemical non-uniformity observed in real deposition processes. Each grid location is simulated with Ngspice independently.

Total Dies Simulated	276
Passing Dies	276
Failing Dies	0
Overall Wafer Yield	**100.0%**
Center Zone Yield	**100.0%**
Edge Zone Yield	**100.0%**
Mean Delay (Wafer)	0.01909 ns
Std Dev Delay (Wafer)	0.000757 ns
Mean Power (Wafer)	97.326 uW

Spatial model: sigma_local = sigma_base x (1 + 2 x r_normalized^2) where r_normalized = r / R_wafer. Edge dies experience 3x higher sigma than center dies, causing yield degradation at the wafer periphery.

Module 5 - PVT Temperature Sweep (-40 deg C to +125 deg C)

Temperature sweep evaluates how carrier mobility degradation and increased leakage at elevated temperature affect propagation delay and power. Each point is a real Ngspice simulation with .TEMP directive.

Temperature Range	-40 deg C to 125 deg C
Min Delay	0.016632 ns at -40 deg C
Max Delay	0.021975 ns at 125 deg C
Delay Range (spread)	**0.005343 ns**
Min Power	96.8646 uW
Max Power	96.8646 uW

Temp (C)	Delay (ns)	Power (uW)	Source	Status
-40	0.016632	96.8646	ngspice	PASS
-20	0.017387	96.8646	ngspice	PASS
0	0.0181	96.8646	ngspice	PASS
27	0.019028	96.8646	ngspice	PASS
50	0.019775	96.8646	ngspice	PASS
75	0.02055	96.8646	ngspice	PASS
100	0.021281	96.8646	ngspice	PASS
125	0.021975	96.8646	ngspice	PASS

Module 6 - Multi-Gate Analysis (Inverter / NAND2 / NOR2)

Each logic gate is simulated independently with real Ngspice SPICE netlists. NAND2 has series NMOS (slower pull-down). NOR2 has series PMOS (slower pull-up). The inverter serves as the baseline reference.

INVERTER	**Delay: 0.019028 ns (1.0x) | Power: 96.86 uW (1.0x) | PASS**
NAND2	**Delay: 0.015279 ns (0.803x) | Power: 96.86 uW (1.0x) | PASS**
NOR2	**Delay: 0.013588 ns (0.714x) | Power: 96.86 uW (1.0x) | PASS**

Multiplier (Nx) shows performance relative to inverter baseline. NAND2 delay is typically 1.3-1.5x inverter. NOR2 is 1.2-1.4x.

Module 7 - Waveform Viewer (Ngspice Transient)

Real V(inp) and V(out) transient waveforms captured from Ngspice using .TRAN 1ps 4ns analysis. Propagation delay extracted from 50% voltage crossings: tpd = (tpHL + tpLH) / 2.

Propagation Delay (tpd)	**0.019028 ns**
Power Dissipation	96.864636 uW
Simulation Source	ngspice
Temperature	27 deg C
VDD/2 Crossing Level	1.65 V

Waveform shows inverting behavior: V(out) is HIGH when V(inp) is LOW and transitions to LOW when V(inp) rises. The 50% crossing offset represents the propagation delay tpd.

Module 8 - Aging Analysis (NBTI + HCI + Electromigration)

NBTI (Negative Bias Temperature Instability) degrades PMOS threshold over time. HCI (Hot Carrier Injection) degrades NMOS near drain. Electromigration MTTF computed via Black's Equation. Models predict how delay and threshold voltage shift over 0-15 years of operation.

Delay at t=0	0.019028 ns
Delay at t=10 yr	**0.019126 ns**
Delay Degradation	0.52%
NBTI Mechanism	PMOS Vth shift (power-law model, Ea=0.13 eV)
HCI Mechanism	NMOS Vth shift (VDD^3 dependence, Ea=0.15 eV)
EM Model	Black's Equation (n=2, Ea=0.9 eV)

Year	dVtn HCI (mV)	dVtp NBTI (mV)	Delay (ns)	Status
0	0.0	0.0	0.019028	PASS
1	30.945	1.564	0.019086	PASS
3	50.734	2.058	0.01911	PASS
5	63.845	2.338	0.019119	PASS
10	87.215	2.781	0.019126	PASS

Module 9 - Power Breakdown Analysis

Total power decomposed into: Dynamic (switching activity alpha*CL*VDD^2*f), Subthreshold leakage (exponential Vth dependence), Gate oxide tunneling (Fowler-Nordheim, tox-dependent), Short-circuit power (crowbar during transition).

Dynamic Power (switching)	57.477091 uW (59.3%)
Subthreshold Leakage	0.000000 uW (0.0%)
Gate Oxide Tunneling	0.000000 uW (0.0%)
Short-Circuit Power	39.387545 uW (40.7%)
Total Power	**96.8646 uW**

Dynamic power dominates in active circuits. Gate oxide tunneling increases exponentially as tox decreases - critical trade-off in sub-5nm nodes.

Module 10 - Sensitivity Tornado Chart

Each parameter is swept from P10 to P90 percentile independently while all others are held at median. The delta delay shows which parameter contributes most to performance variation.

Rank	Parameter	Delay @P10 (ns)	Delay @P90 (ns)	Delta (ns)	Impact %
#1	load_cap	0.01673	0.021232	0.004502	21.2%
#2	length	0.018018	0.019991	0.001973	9.9%
#3	tox	0.018468	0.019564	0.001096	5.6%
#4	width_p	0.01951	0.018604	0.000906	4.6%
#5	width_n	0.019482	0.018621	0.000861	4.4%
#6	vth_n	0.018746	0.019278	0.000532	2.8%

Parameters with the longest bars require tightest manufacturing control. load_cap and channel length dominate delay variation in 180nm CMOS.

Summary and Conclusion

PyYield is an open-source Python-based Electronic Design Automation (EDA) platform for CMOS yield prediction and process variation analysis. It integrates real Ngspice transient simulation, Monte Carlo LHS sampling with Pelgrom mismatch, Random Forest ML surrogate modeling, wafer spatial analysis, aging reliability prediction, and interactive visualization into a complete variability-aware design workflow.

Overall Monte Carlo Yield: 93.67%

ML Oracle Yield Estimate: 70.0%

Key findings: Threshold voltage (Vth_n) and channel length (L) are the dominant contributors to delay variation as confirmed by SHAP analysis and the tornado sensitivity chart. Edge dies on the 300mm wafer show higher delay spread due to radial process variation gradients. The Slow-Slow corner represents the critical timing path and must be verified against spec limits. Aging analysis confirms less than 1% delay increase over 10 years of operation under NBTI and HCI stress at nominal conditions.

Disclaimer: Results are based on LEVEL-1 MOSFET models and are suitable for educational and research purposes at 180nm-500nm nodes. For tape-out decisions at sub-65nm, BSIM4 or FinFET models with foundry PDKs are required. PyYield is not affiliated with any semiconductor foundry.